Equations of Parallel & Perpendicular Lines — Notes

Watch the video and fill in each blank. Pause when you see “Your turn.”

Name: _____ Date: _____

☞ *Video: Slope rules*

Key Idea

_____ lines have the _____ slope.

_____ lines have slopes that are _____.

Flip the fraction **and** change the _____: $\frac{2}{3} \to$ _____, and $\frac{2}{3} \cdot \left(-\frac{3}{2}\right) =$ _____.

☞ *Video: Find the slopes*

Watch Me — Worked Example

$y = \frac{2}{3}x + 1$: slope _____ parallel slope _____ perpendicular slope _____

☞ *Video: Write a parallel line*

Watch Me — Worked Example

Through $(2, 3)$, parallel to $y = 4x - 1$.

same slope: $m =$ _____

$y - 3 = 4(x - 2)$ _____ form

$y =$ _____

☞ *Video: Write a perpendicular line*

Watch Me — Worked Example

Through $(4, 1)$, perpendicular to $2x + 3y = 6$.

solve for y: $y = -\frac{2}{3}x + 2$ slope _____

perpendicular slope _____

$y - 1 = \frac{3}{2}(x - 4)$ $y =$ _____

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☞ *Video: Watch out*

Common Mistake

Perpendicular needs **both** steps: flip **and** change the sign. The perpendicular slope of 2 is _____, not $\frac{1}{2}$.

☞ *Video: Your turn*

Your Turn

Write the line through $(3, -2)$ perpendicular to $y = -3x + 7$.

Slope _____: $y + 2 = \frac{1}{3}(x - 3) \Rightarrow y =$ _____

Notice

Remember: Parallel: same slope. Perpendicular: negative reciprocal — flip and change the sign. Then use point-slope form $y - y_1 = m(x - x_1)$.

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